Chapter 3 — Finite elements in d dimensions

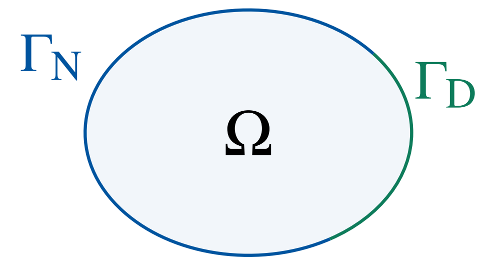
The reference element, polynomial spaces and the geometrical map

MTH3340 · Numerical Methods for PDEs

1 The boundary value problem in d dimensions

Let $\Omega \subset \mathbb{R}^d$ ($d = 2, 3$) with boundary $\Gamma \doteq \partial\Omega$ split into a Dirichlet part Γ_D and a Neumann part Γ_N.

$$\begin{cases} \mathcal{A}\, u(\boldsymbol{x}) = f(\boldsymbol{x}) & \text{in } \Omega, \\ \mathcal{B}_\mathrm{D}\, u(\boldsymbol{x}) = u_\mathrm{D}(\boldsymbol{x}) & \text{on } \Gamma_\mathrm{D}, \\ \mathcal{B}_\mathrm{N}\, u(\boldsymbol{x}) = g_\mathrm{N}(\boldsymbol{x}) & \text{on } \Gamma_\mathrm{N}. \end{cases}$$



$\mathcal{A}$ is a differential operator (e.g. $-\Delta$), $\mathcal{B}_\mathrm{D}$ a trace operator and $\mathcal{B}_\mathrm{N}$ a flux operator.

Nothing here is new in kind — only in d. The whole difficulty moves to the construction of the discrete space.

The weak form

Multiply by a test function, integrate over Ω and integrate by parts. Dirichlet data are essential (built into the space); Neumann data are natural (they land in the right-hand side).

Example (Poisson problem with mixed conditions)

$-\nabla \cdot \boldsymbol{\kappa}\nabla u = f$, $u = u_{\mathrm{D}}$ on Γ_{D}, $\boldsymbol{n} \cdot \boldsymbol{\kappa}\nabla u = g_{\mathrm{N}}$ on Γ_{N}. Pick any extension $Eu_{\mathrm{D}} \in H^1(\Omega)$ with $Eu_{\mathrm{D}} = u_{\mathrm{D}}$ on Γ_{D}. Find $u_0 \in H^1_{\Gamma_{\mathrm{D}}}(\Omega)$ with

$$\int_\Omega \boldsymbol{\kappa}\nabla u_0 \cdot \nabla v = \int_\Omega f v + \int_{\Gamma_{\mathrm{N}}} g_{\mathrm{N}} v - \int_\Omega \boldsymbol{\kappa}\nabla[Eu_{\mathrm{D}}] \cdot \nabla v \qquad \forall v \in H^1_{\Gamma_{\mathrm{D}}}(\Omega),$$

and set $u \doteq u_0 + Eu_{\mathrm{D}}$.

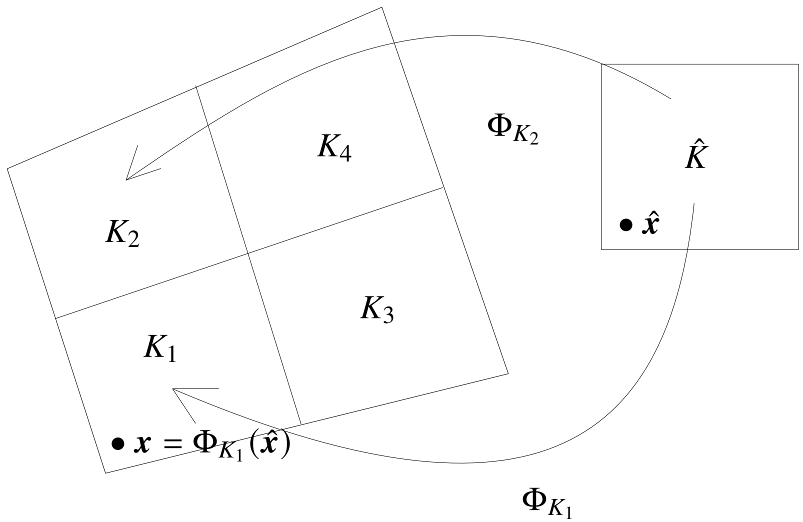
This is the offset function method of Chapter 2, verbatim. Abstractly: find $u \in X$ with $a(u, v) = \ell(v)$ for all $v \in X$.

Galerkin, and the one thing we are missing

Find $u_h \in X_h$ such that $a(u_h, v_h) = \ell(v_h)$ for all $v_h \in X_h$, with $X_h \subset X$ conforming. Céa's lemma still applies. So: how do we build X_h in 2D and 3D?

In 1D we wrote the hat functions down explicitly on a partition of (a, b). In d dimensions there is no such global formula. Instead, everything is built cell by cell from one reference cell $\hat{K}$:

1. build a space on $\hat{K}$;
2. push it to each K through $\mathbf{\Phi}_K$;
3. assemble, enforcing C^0.

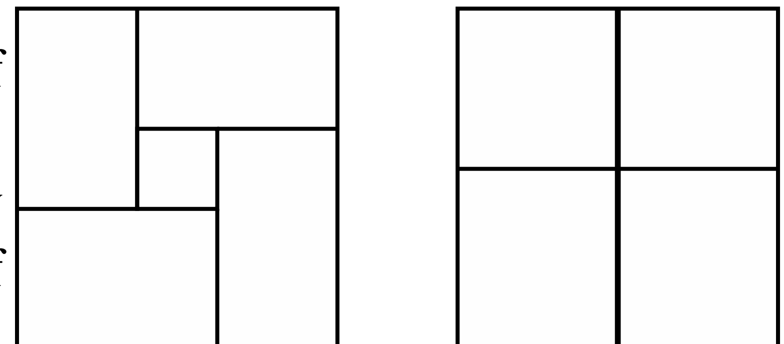


2 The mesh

Definition (Conforming triangulation)

A partition $\mathcal{T}_h$ of Ω into cells K is *conforming* if for any two neighbours K^+, K^- the intersection $\overline{K}^+ \cap \overline{K}^-$ is either empty or a whole k-face ($k < d$) of both cells.

A k-face of a polytope: the cell itself is a d-face, then faces, edges, vertices. Hanging nodes are excluded — they would break the gluing of degrees of freedom.



Non-conforming Conforming

Every $K \in \mathcal{T}_h$ carries a reference cell $\hat{K}$ and a diffeomorphism $\mathbf{\Phi}_K : \hat{K} \to K$; write $\hat{\boldsymbol{x}} \doteq \mathbf{\Phi}_K^{-1}(\boldsymbol{x})$.

3 The reference finite element

Definition (Finite element (Ciarlet))

A finite element is a triplet $\{K, \mathcal{V}, \Sigma\}$ where K is a compact, connected, Lipschitz subset of $\mathbb{R}^d$, $\mathcal{V}$ is a vector space of functions on K, and $\Sigma = \{\sigma_1, \dots, \sigma_{n_\Sigma}\}$ is a set of linear functionals forming a basis of the dual space $\mathcal{V}'$.

The σ_i are the degrees of freedom. Because Σ is a dual basis, there is a unique basis $\{\phi^1, \dots, \phi^{n_\Sigma}\}$ of $\mathcal{V}$ with

$$\sigma_i(\phi^j) = \delta_{ij},$$

the shape functions: one per degree of freedom. We build the triplet first on the reference cell, $(\hat{K}, \hat{\mathcal{V}}, \hat{\Sigma})$.

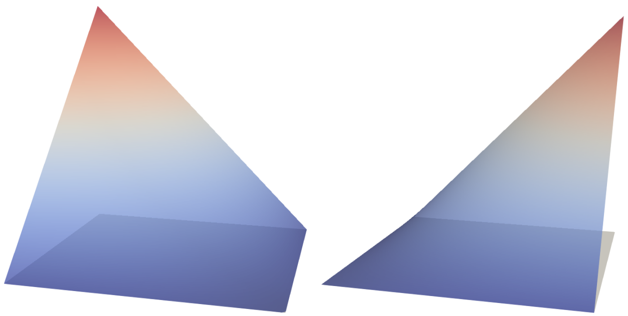
Example: the bilinear reference element

$\hat{K} \doteq [-1, 1]^2$, and

$$\begin{aligned} \hat{\mathcal{V}} &= \mathcal{P}_1(\hat{x}) \otimes \mathcal{P}_1(\hat{y}) = \{1, \hat{x}\} \otimes \{1, \hat{y}\} \\ &= \operatorname{span}\{1, \hat{x}, \hat{y}, \hat{x}\hat{y}\} \doteq \mathcal{Q}_1, \end{aligned}$$

with $\hat{\Sigma} = \{\hat{\sigma}_i(\hat{v}) \doteq \hat{v}(\hat{\boldsymbol{x}}_i)\}$, the evaluations at the four vertices.

Is $\hat{\Sigma}$ a basis of $\hat{\mathcal{V}}'$? Two bilinear polynomials with the same four nodal values are equal, so $\hat{v} \mapsto (\hat{\sigma}_1(\hat{v}), \ldots, \hat{\sigma}_4(\hat{v}))$ is injective between spaces of the same dimension — hence bijective. This is unisolvence.



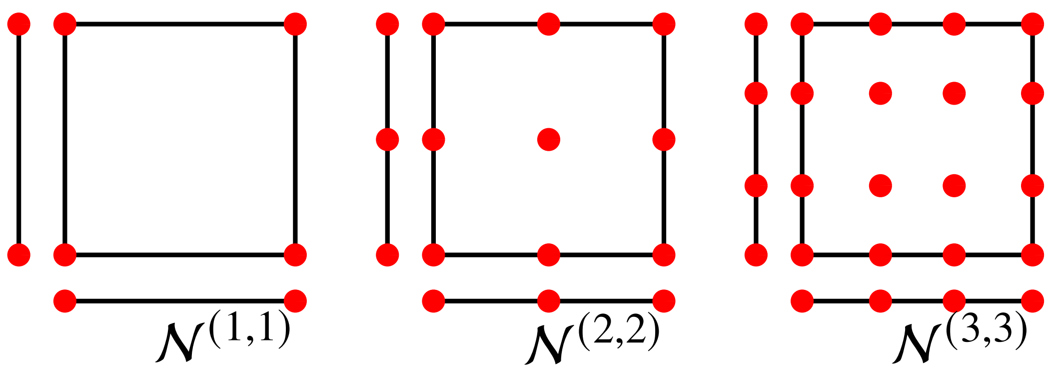
Two shape functions of $\mathcal{Q}_1$ on $\hat{K}$.

Lagrangian bases and tensor-product nodes

Given nodes $\{x_0, \ldots, x_k\}$ in $\mathbb{R}$, the 1D Lagrange polynomials are

$$\ell_m^k(x) \doteq \frac{\prod_{n\neq m}(x - x_n)}{\prod_{n\neq m}(x_m - x_n)}, \qquad \ell_m^k(x_l) = \delta_{ml}.$$

On a d-cube we take tensor products, both of the nodes, $\mathcal{N}^{\boldsymbol{k}} = \mathcal{N}^{k_1}\times\cdots\times\mathcal{N}^{k_d}$, and of the polynomials, $\ell_{\boldsymbol{m}}^{\boldsymbol{k}}(\boldsymbol{x}) \doteq \prod_{i=1}^{d} \ell_{m_i}^{k_i}(x_i)$. E.g. in 2D for $p = 1$: $\hat{b}^1 = \ell_0(\hat{x})\ell_0(\hat{y})$, $\hat{b}^2 = \ell_1(\hat{x})\ell_0(\hat{y})$, $\hat{b}^3 = \ell_0(\hat{x})\ell_1(\hat{y})$, $\hat{b}^4 = \ell_1(\hat{x})\ell_1(\hat{y})$. So $\dim Q_p = (p+1)^d$ and the shape functions come for free.

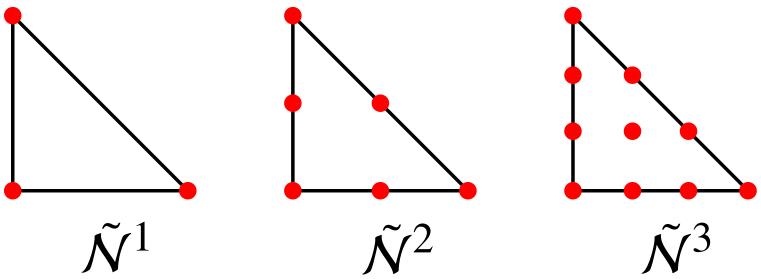


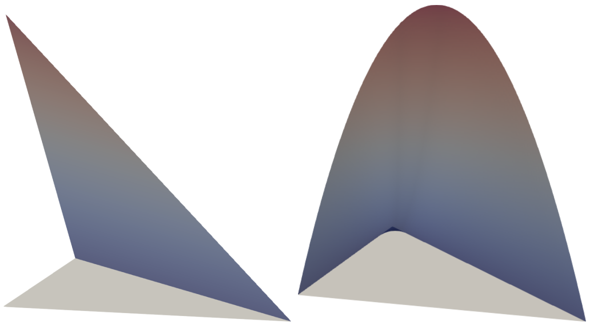
Simplices: the spaces $\mathcal{P}_p$

On triangles and tetrahedra the tensor product is too rich. We truncate:

$$\mathcal{P}_p \doteq \operatorname{span}\{\boldsymbol{x}^{\boldsymbol{\alpha}} \;:\; |\boldsymbol{\alpha}| \le p\}, \qquad |\boldsymbol{\alpha}| \doteq \textstyle\sum_i \alpha_i.$$

In 2D, $\mathcal{P}_1 = \{1, x, y\}$ (dim 3) and $\mathcal{P}_2 = \{1, x, y, xy, x^2, y^2\}$ (dim 6); in general $\dim \mathcal{P}_p = \binom{p+d}{d}$. The nodes are truncated the same way, $\tilde{\mathcal{N}}^p = \{\boldsymbol{s} \in \mathcal{N}^{p\mathbf{1}} : |\boldsymbol{s}| \le p\}$ — vertices, then edge midpoints, and so on.





Shape functions of $\mathcal{P}_1$ and $\mathcal{P}_2$.

Shape functions from a pre-basis

On simplices there is no tensor-product shortcut, and for high p the formulas get unpleasant. A code should generate them. Take any pre-basis $\{\psi^1, \ldots, \psi^{n_\Sigma}\}$ of $\mathcal{V}$ — monomials will do — and look for $\phi^i = \Xi_{ij}\psi^j$.

Imposing $\sigma_k(\phi^i) = \delta_{ik}$ and writing $\mathbf{C}_{kj} \doteq \sigma_k(\psi^j)$,

$$\delta_{ik} = \sigma_k(\Xi_{ij}\psi^j) = \Xi_{ij}\,\sigma_k(\psi^j) = \Xi_{ij}\mathbf{C}_{kj} \quad\Longrightarrow\quad \mathbf{C}\,\boldsymbol{\Xi}^T = \mathbf{I}, \qquad \boldsymbol{\Xi} = \mathbf{C}^{-T}.$$

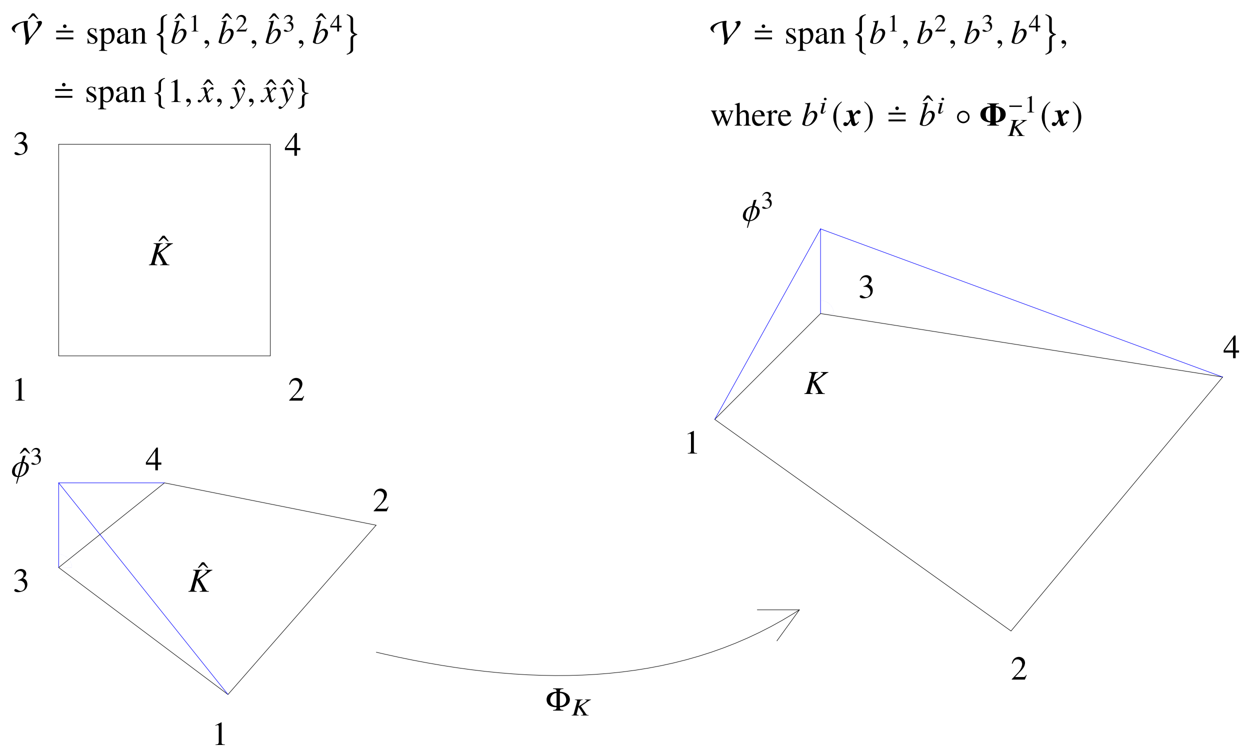
One small dense solve, once, on the reference cell. It also tells us when the element is well defined: $\mathbf{C}$ is invertible exactly when Σ is unisolvent on $\mathcal{V}$.

4 From the reference to the physical cell

Given $(\hat{K}, \hat{\mathcal{V}}, \hat{\Sigma})$ and the map $\mathbf{\Phi}_K$, the finite element $(K, \mathcal{V}, \Sigma)$ on the physical cell is

$$K \doteq \mathbf{\Phi}_K(\hat{K}), \qquad \mathcal{V} \doteq \{\hat{v} \circ \mathbf{\Phi}_K^{-1} \,:\, \hat{v} \in \hat{\mathcal{V}}\}, \qquad \Sigma \doteq \{\sigma_i(v) \doteq \hat{\sigma}_i(v \circ \mathbf{\Phi}_K)\}.$$

We never build a space on K from scratch: everything is $\hat{K}$ plus a map.



Degrees of freedom and shape functions on K

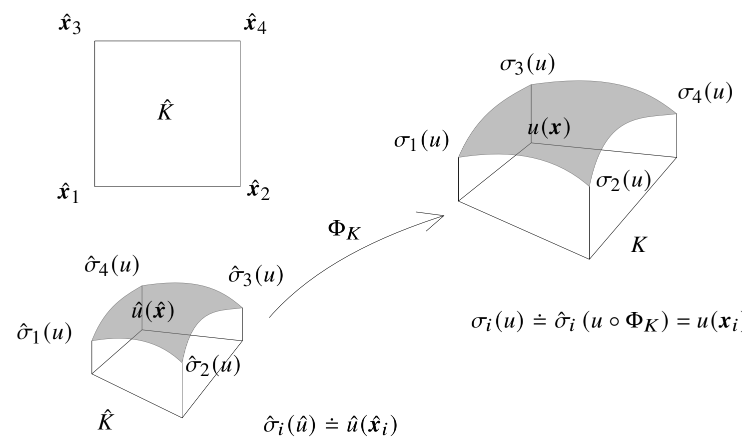
For Lagrangian elements the pull-back is transparent,

$$\sigma_i(v) = \hat{\sigma}_i(v \circ \mathbf{\Phi}_K) = v(\boldsymbol{x}_i),$$

just the nodal value at the mapped node. And the duality survives the map: with $\phi^j \doteq \hat{\phi}^j \circ \mathbf{\Phi}_K^{-1}$,

$$\sigma_i(\phi^j) = \hat{\sigma}_i(\hat{\phi}^j) = \delta_{ij}.$$

So the local interpolator $\pi_K(v) \doteq \sum_i \sigma_i(v)\, \phi^i$ is a projection onto $\mathcal{V}$.



5 How do we compute Φ_K?

The mesh gives us the vertex positions $\boldsymbol{x}^i$ of each cell. Use the finite element machinery on the geometry itself:

$$\boldsymbol{x} = \boldsymbol{\Phi}_K(\hat{\boldsymbol{x}}) \doteq \sum_i \hat{\phi}_1^i(\hat{\boldsymbol{x}})\,\boldsymbol{x}^i,$$

with $\{\hat{\phi}_1^i\}$ the *linear* reference shape functions. Since $\hat{\phi}_1^i(\hat{\boldsymbol{x}}_j) = \delta_{ij}$, the map sends each reference vertex to the corresponding physical vertex, as required.

Triangle ($\hat{\mathcal{V}} = \mathcal{P}_1$): three vertices determine $\boldsymbol{\Phi}_K = \mathbf{A}\hat{\boldsymbol{x}} + \boldsymbol{b}$, affine, so $\boldsymbol{J}_K = \mathbf{A}$ is *constant* on the cell and $\boldsymbol{\Phi}_K^{-1}(\boldsymbol{x}) = \mathbf{A}^{-1}\boldsymbol{x} - \mathbf{A}^{-1}\boldsymbol{b}$.

Quadrilateral ($\hat{\mathcal{V}} = \mathcal{Q}_1$): the $\hat{x}\hat{y}$ term survives, so $\boldsymbol{\Phi}_K$ is bilinear, not affine, and $\boldsymbol{J}_K$ varies within the cell.

Which space do we actually get on K?

Triangles. $\hat{v} = a+b\hat{x}+c\hat{y}$ and $\hat{\boldsymbol{x}} = \tilde{\mathbf{A}}\boldsymbol{x}+\tilde{\boldsymbol{b}}$ is affine, so

$$v = \hat{v}\circ\boldsymbol{\Phi}_K^{-1} = \tilde{a}+\tilde{b}x+\tilde{c}y \in \mathcal{P}_1(K). \quad \checkmark$$

Affine maps preserve $\mathcal{P}_p$.

General quadrilaterals. $\hat{v} \in \mathcal{Q}_1(\hat{K})$ but $v = \hat{v} \circ \boldsymbol{\Phi}_K^{-1} \notin \mathcal{Q}_1(K)$ — the map is not affine, so it does not preserve the space. Equality holds only when K is a parallelogram (stretch + rotation + translation of $\hat{K}$).